

Yasemin Ozkut

MS Candidate | Seeking Full-Time AI/ML Roles (May 2026)

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Education

MS	The Ohio State University (OSU) , Electrical and Computer Engineering • Focus: Multimodal Learning, VLMs, Computer Vision	Columbus, OH, USA Aug 2024 – Expected May 2026
BS	Sabanci University , Computer Science and Engineering • Minor: Business Analytics, School of Management	Istanbul, Turkey Sept 2019 – Feb 2024

Skills

Languages: Python (3+ years hands-on, 1+ year professional), C++, C, C#, SQL, JavaScript

Frameworks & Libraries: PyTorch, Pytorch Lightning, CUDA, YOLO, Keras-OCR, OpenCV, Unsloth, Hugging Face, Monai, Hydra, LangChain, LangGraph, Pandas, NumPy, Scikit-learn, React.js, Node.js, Express.js, MongoDB, Flutter

Tools: Git, SLURM, Command Line (Linux), LaTeX, Gephi, Postman, Unity, Figma

Techniques: Deep/Machine Learning, Multimodal Vision-Language Models (VLMs/LLMs), Distributed Training, Prompt Engineering, AI-Assisted Coding, AI Agents, Fine-Tuning, Hyperparameter-Tuning, Computer Vision, OOP

Publications

ERDES: A Benchmark Dataset for Retinal Detachment Classification in Spatiotemporal Ocular Ultrasound , <i>Nature Scientific Data</i> 🔗 Dataset Website Hugging Face Pouyan Navard, Yasemin Ozkut , Sirikar Adhikari, Alper Yilmaz	In revisions July 2025
HOPE: A Benchmark Dataset for Pulmonary Embolism Classification in Spatiotemporal Heart Ultrasound , <i>Nature Scientific Data</i> Yasemin Ozkut , Pouyan Navard, Sirikar Adhikari, Alper Yilmaz	Under preparation Sept 2025 - Present

Experience

Ubihere Computer Vision Engineer Intern (Full-Time)	Aug 2025 – Present Columbus, OH
• Wrapped research-stage person re-identification modules into a configurable, GPU-accelerated Flask microservice with REST endpoints and a clean JSON API, which is now being integrated into the company's core product. • Reduced inference latency by up to 67% through GPU batching, NumPy-vectorized similarity computation, parallelized image I/O, and adaptive workload routing. • Containerized the service with Docker and NVIDIA Container Toolkit, implementing health checks, structured logging, and resource controls for production deployment. • Currently developing an AI-powered admin dashboard that adapts to customer camera environments and delivers tracking insights.	
Photogrammetric Computer Vision Lab (PCVLab) Graduate Research Assistant	Aug 2024 – Aug 2025 Columbus, OH
• ERDES Benchmark and Publication 🔗 • Developed a two-stage diagnostic pipeline for retinal detachment classification in ocular ultrasound, achieving 0.94–0.99 (Non-Retinal Detachment vs. Retinal Detachment) and 0.73–0.88 (Macula intact vs. detached). • Trained and compared 16 classifiers across 8 deep 3D CNN/transformer architectures (UNETR, SwinUNETR, V-Net, UNet++, SNet154, 3D ResNet, and 3D UNet), enabling systematic benchmarking. • Extended a Lightning-Hydra pipeline for stability, hyperparameter tuning, and distributed training, enabling systematic experimentation.	

• **Multimodal Learning for Chest X-Ray Interpretation**

- Curated and preprocessed MIMIC-CXR chest X-rays dataset from 377K images / 228K studies to 100K images / 50K studies, pairing frontal-lateral views with structured findings and impressions.
- Built and fine-tuned a multi-modal chest X-ray VLM (ViT encoder + GPT/LLaMA 3.2 via Unsloth) for joint image-report understanding and automated report generation.
- Implemented custom dataset classes, masked patching, and scaled multi-GPU training on HPC clusters with Hugging Face Accelerate, CUDA, and SLURM.

DAI-Labor

July 2023 – Sept 2023

AI/ML Researcher Intern (supported by Erasmus+ mobility program)

Berlin, Germany

- Built a real-time YOLOv8-based multi-object detection model for German supermarket products, achieving 87.2% test accuracy.
- Collected and annotated 1.2K product images, expanding to 3.5K with augmentations based on the Freiburg Groceries Dataset using Roboflow.
- Integrated OCR (Keras-OCR) for German-language labels and conducted evaluation of both YOLOv8 and OCR models, comparing performance using F1 scores, confusion matrices, and loss metrics.

Projects

Semantic Person Re-Identification (ReID) via LangGraph and Vision-Language

July 2025

Reasoning, Personal Project 

- Built a LangGraph pipeline integrating YOLOv8, Qwen2.5-VL-3B, and Qwen2.5-7B for person detection, description generation, and identity assignment across frames/videos.
- Designed structured JSON prompts to extract rich semantic attributes (hair, clothing, accessories, face shape) and implemented persistent memory for global ID tracking.
- Enabled explainable matching with logged reasoning, confidence scores, and memory updates for re-identification.

Serious Game for Children with Cerebral Palsy, Sabanci University

Feb 2023 – Feb 2024

- Developed a serious game in Unity using C# for balance physiotherapy sessions in children with Cerebral Palsy.
- Collected live data from Wii Balance Board for each game session and stored it into MongoDB.
- Generated visual reports from the processed data using Python and Flask.

Spotify Music Artist Success Collaboration Network, Sabanci University

Feb 2023 – June 2023

- Collected and preprocessed Spotify and Kaggle data to build a collaboration network of 8.6K artists and 13.3K connections using Python/NetworkX, incorporating genre, popularity, and centrality metrics.
- Applied clustering and centrality algorithms to identify key influencers, and visualized the network with Gephi for success analysis.

Agile Full Stack Software Development Project, Sabanci University

Oct 2022 – Jan 2023

- Built a full-stack MERN web application for the Turkish Football Federation with data visualization, referee assignment, and improved GUI design.
- Practiced modern software engineering workflows including version control, scrum, automated testing (Selenium), and API documentation (Swagger).

Membership & Volunteer Work

Hiphop Dancer (2015–Present); **Board Member**, SuDance Club, Sabancı University (2019–2023); **Contestant**, Peak Games Unithon (2022); **Tutor**, Civic Involvement Project – 4th-grade students (2020)